Will Kilgore

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Education

McGill University **Expected Dec 2027**
Bachelor of Engineering in Mechanical Engineering - GPA: 3.84/4.0 *Montreal, QC*

Work Experience

Automated Tire, Inc. **May 2026 – Aug 2026**
Mechanical Engineering Intern – Autonomous tire-changing robotic platform *Woburn, MA*

- Designed, fabricated, and validated a custom life-cycle testing jig, sourcing components and supplier quotes to evaluate tire-rotation mechanism durability under repeated loading
- Created technical drawings and assembly documentation in Onshape for 100+ component production assemblies, enabling technician-led builds without engineer oversight and reducing assembly time
- Designed, fabricated, and iterated process-specific fixtures and tooling, selecting manufacturing processes based on cost, lead time, and production volume to resolve assembly issues and reduce production time by 40%
- Simplified top-level assembly CAD by implementing a fastener-visibility toggle, reducing mate solving time from 3.5s to 1.2s (66% faster) across 4,000+ fastener instances, improving CAD responsiveness for design team
- Identified and documented design discrepancies during robotic system assembly, provided structured redline feedback, and processed engineering change orders that resolved recurring build issues across 2 design iterations
- Maintained three multi-material FDM printers, including calibration, hardware servicing, and failure diagnosis, reducing print failures and achieving same-day prototype turnaround for engineering team

Parlee Composites **June 2025 – Present**
Manufacturing and Engineering Intern – Carbon fiber bicycle design and manufacturing *Beverly, MA*

- Conducted experimental vibration analysis using accelerometer data and analytical models to evaluate bicycle frame compliance, informing design decisions for a new frame platform
- Developed a graphical fit-analysis system overlaying generated rider fit coordinates with 200 existing fits, reducing required production part count by 25% while maintaining rider coverage.
- Implemented an Excel-based fit analysis tool generating 150+ stem stack/reach coordinate combinations across varying stem lengths, stem angles, and frame sizes to reduce dealer consultation time
- Introduced new prepreg composite layups that reduced void content and post-processing time, increasing small-part production throughput by up to 30%
- Managed quality control for overseas frame production using inspection checklists and defect tracking, reducing recurring supplier issues through improved defect identification and feedback loops

Projects

McGill Rocket Team **Sept 2023 – Present**
Airframe Composites Project Co-Lead

- Co-led composites manufacturing for the aerostructures subteam, coordinating layups, tooling, and assembly integration across a 10-member subteam to deliver flight-ready structures for Launch Canada
- Designed and fabricated molds for a 42” composite nose cone as part of the aerostructures subteam, enabling the transition to a fully student-built 8” rocket platform
- Executed vacuum-assisted resin infusion (VARI) and prepreg layups to manufacture lightweight, high-strength structural body tubes for competition rockets

Skills

Technical Skills: Rapid Prototyping (3D Printing), Design for Manufacturing & Assembly (DFMA), GD&T, FEA, Vibrational Analysis, Experimental Design, Composite Manufacturing (Prepreg, VARI), Cable Routing & Harnessing

Software: CAD (SolidWorks, Onshape), Lifecycle Management (Arena), FEA (Abaqus), Python, Excel, AI-assisted workflows

Relevant Courses: Machine Element Design, Mechanics of Deformable Solids, Principles of Manufacturing, Dynamics, Numerical Methods, Materials Engineering, Fluid Mechanics 1&2, Thermodynamics 1&2

McGill Cycling Team **Aug 2023 – Present**
VP of Finance
- Coordinated logistics and safety for weekly team training and social rides, managing groups of 15+ riders
- Managed fundraising, sponsorships, and team fees to sustain trip subsidies and financial support

Potential ATI Bullets:
- Generated fabrication-ready CAD and drawings for outsourced prototype manufacturing
- Something about sheet metal?

Other Skills: Tooling and fixture design

Projects

Carbon Fiber Bike Components **May 2025 – Aug 2025**
- Designed and developed custom carbon fiber bicycle components using forged carbon fiber and custom molds to meet specific geometry and weight constraints
- Used SolidWorks, Cura, and 3D printing for rapid prototyping to validate component geometry and reduce iteration time

- Performed FEA in Abaqus and iterated designs based on simulation and physical fit validation to optimize strength-to-weight performance

DFMA
SOlidworks
Onshape
Abaqus
Numerical Analysis
Python
FEA
Design of experiments
Data analysis
Vibrational analysis
Excel
Data visualization
Team coordination
Prepreg
Vari
Cable routing and harnessing
Lab skills
gd&T
Composites
3d printing
Aerostructures
Shop tools
Laser engraving
Thermodynamics
Graphic design

Fluid mechanics
Rapid prototyping
Quality control
Manufacturing
Manufacturing process improvement
Tooling and fixture design
Prototype development
Electromechanical systems
Engineering documentation
Technical drawings
CAD
Product development